

Tobi Bui

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Education

North Carolina State University
Master of Financial Mathematics
GPA: 4.0/4.0

Raleigh, NC
December 2026

Oberlin College
Bachelor of Arts and Science in Computer Science and Economics
GPA: 3.6/4.0

Oberlin, OH
May 2025

Related Coursework

Options & Derivatives Pricing, Linear Optimization, Fundamentals of Statistical Inference, Fixed Income Product & Analysis, Machine Learning in Finance, Bayesian Computation, Akuna Capital Option 101

Skill

Programming: Python (Pytorch/Tensorflow, Selenium, Pandas/Numpy, Scikit-Learn), R, SQL (DuckDB, MongoDB)

Technology: Microsoft Office Suite, Bloomberg Terminal, Linux, SAS Enterprise, Tableau, Power BI

Experience

Oberlin College
Research Assistant

Oberlin, OH
06/2024 – 05/2025

- Engineered predictive models in Python (neural networks, ensemble algorithms, and survival-specific regression models) to estimate patient survival probabilities, achieving a C-Index of 0.75 for model performance
- Applied survival analysis techniques that include Log-Rank test, survival regression, and Cox models to analyze time-to-event data, highlighting strong capabilities in hazard modeling, time-dependent covariates, and inferential statistics.
- Utilized Mahalanobis distance matching to mitigate selection bias in imbalanced case-control studies, assessed treatment effects through ATE estimation, SMD diagnostics, and non-parametric Wilcoxon tests to ensure statistical validity.
- Presented poster at Penn Conference on Big Data in Biomedical and Population Health Sciences as the only undergraduate student

VinBigData
Machine Learning Engineer Intern

Hanoi, Vietnam
05/2023 – 08/2023

- Built a customized crawler bot to retrieve data from more than 100 sources to train Vivi, a VinFast AI chat-bot
- Trained Q/A, text summarizing tasks in Vietnamese data using Google MT5-small pre-trained model, resulting in 50% higher accuracy and better response compared to other models
- Smoothed the models by fine-tuning and hyperparameter-tuning for specific tasks, increased the model's efficiency by 30%, and decreased GPU power by 15%

Project Experience

Quant Trading Project | Streamlit, Pytorch, Alpaca, Gurobi, [Git](#)

12/2023 - Present

- Deployed a Pair-Trading strategy with Kalman Filter and Meta Prophet Time Series to forecast future prices and trade based on futuristic prices, estimating earning around 20% profit with backtesting and optimization
- Linearized convex Quadratic Problem of MPT with Gurobi Python's API and skfolio for portfolio optimization of maximizing returns and reduce risks, generating 15% increase in profits for investment strategies stored in DuckDB, queried with SQL
- Automated Financial Analysis for future evaluation with SEC EDGAR HTTP requests to conduct Equity Options Trading

MBS Credit Risk Modelling & Trading Strategies | Alpaca, Stochpy, Scikit-learn, Streamlit

08/2025 – 12/2025

- Built MBS cash flow modeling and pricing under stochastic economic scenarios with Python
- Apply logistic regression and Cox models for default prediction, PSA and Vasicek models for prepayments, and Monte Carlo simulations for valuation
- Built a quantitative trading platform on Streamlit with a backtest framework and actionable insights into bond market investments

Corporate Spread Prediction | PyTorch, Scikit-learn, Statsmodels, [Git](#)

02/2025 – 05/2025

- Customized simple feedforward neural network (SFNN) model to predict corporate bond spreads using data from FRED API, achieving an R^2 of 0.7 for model credibility of detecting patterns on future recessions.
- Cleaned data with feature engineering of VIF, Kalman Filter, and Yeo-Johnson transformation to fit SFNN's learning process
- Applied Weight Least Squares regression for analyzing economic variables' impact on Corporate Spread and tuned the weight for SFNN, forcing the model to be realistic